Note on Ethics

Ethical Dimensions of Collecting, Storing & Querying Reddit Data

Subreddit: r/cscareerquestions | System: RAG Pipeline | April 2026

(i) Personal Information & Re-Identification Risk

Reddit usernames are persistent pseudonyms rather than true anonymizers. Over time, a single user posting across multiple threads can — without ever revealing their legal name — assemble a profile that is highly identifying when viewed in aggregate. This is a well-documented privacy risk known as the aggregation problem: individually harmless data points combine to create a fingerprint that can narrow a person's identity to a very small set of candidates.

Speculative Scenarios from This Dataset

I want to be transparent: with a corpus of 15,000+ posts, it is not feasible to manually read through every thread to surface concrete re-identification cases. I did not find a specific confirmed instance during this project. However, based on the nature of discussions in r/cscareerquestions, I can speculate with reasonable confidence about the kinds of combinations that could occur:

> **Illustrative Example (Speculative):**
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> Suppose a user posts across four different threads over six months:
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> - **Thread A:** "I work at [named mid-size fintech startup], L4 SWE, two years in"
> - **Thread B:** "Graduated from [named state university] in 2021 with a CS degree"
> - **Thread C:** "On OPT, applying for H1-B this cycle — nervous about the lottery"
> - **Thread D:** "Got a competing offer of $145k base in [named city], should I counter?"
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> No single post is identifying. But combining all four — employer, graduation year, visa status, salary range, and city — narrows the candidate pool dramatically, potentially to a handful of individuals. Cross-referencing with LinkedIn or GitHub could complete the identification. Discussions of this kind are routine in r/cscareerquestions, making this scenario plausible across many users in the corpus.

Other plausible risk combinations include: a user describing a very specific workplace incident at a named company alongside their job title and tenure (which could identify them to their employer); or a user disclosing a mental health condition alongside career history in a way that could be traced back with knowledge of their professional network.

High-Level Detection Methods (Not Implemented)

Given the scale of the corpus, finding such cases computationally would require dedicated tooling. The following approaches could be implemented in future work, though they represent significant engineering effort that was out of scope for this project:

- **Cross-post aggregation analysis:** Group all posts by username, then run named entity recognition (NER) to extract employer names, locations, universities, visa types, and salary figures per user. Flag users where the combination of extracted entities creates a high-specificity profile.
- **Quasi-identifier detection:** Apply k-anonymity or l-diversity analysis to the set of (location, employer, graduation year, visa status) tuples; any combination with $k < 5$ represents a re-identification risk.
- **Cross-platform linkage simulation:** Attempt to match high-risk usernames against public LinkedIn or GitHub data programmatically — not to actually identify people, but to measure how many matches can be made, as a risk metric.

- **PII scrubbing at ingestion:** Use an NER model at the point of data collection to redact or pseudonymize high-risk entities (salaries, visa statuses, employer names) before writing to the database.

These methods would meaningfully reduce re-identification risk but require careful implementation and would themselves involve additional privacy-sensitive processing. They are flagged here as recommended future work for any production deployment of this system.

(ii) The Right to Be Forgotten

The Right to Be Forgotten — codified in GDPR Article 17 and reflected in emerging data protection frameworks globally — grants individuals the right to request deletion of their personal data from systems that hold it. Reddit partially honours this: when a user deletes a post, it is removed from Reddit's platform and no longer served by the API. Our RAG pipeline, however, creates a secondary copy of the data in a local database, entirely outside Reddit's deletion machinery. In its current form, the system does not comply with this right.

The practical consequence is significant. If a user posts a vulnerable disclosure — say, a candid description of being laid off, struggling with mental health, or negotiating a job offer — and later deletes that post, our system still holds it. The deleted text can still be retrieved by the vector store, surfaced as context to the language model, and quoted in answers to future users. From the posting user's perspective, they exercised their right to remove their words from the internet. From our system's perspective, nothing changed.

Achieving full compliance in a production RAG system is technically feasible but requires deliberate upfront engineering. It would involve maintaining a post ID-to-chunk-to-embedding mapping so that deletions can cascade through all derived representations; running a periodic re-sync against the Reddit API to detect posts that have since been deleted or marked [removed]; and explicitly deleting the corresponding embedding vectors from the vector store. Most modern vector stores (Chroma, Pinecone, Weaviate) support deletion by ID, so the mechanism is available — it simply was not built into this prototype.

The honest answer to whether full compliance is realistic for a production system is: yes, but only if it is treated as a first-class engineering requirement from the start. Retrofitting deletion compliance into an existing pipeline — where chunk IDs were not tracked, or embeddings were stored without provenance — is significantly harder and may require a full re-ingestion. The practical lesson is that Right-to-Be-Forgotten compliance must be a design decision, not an afterthought. For this academic prototype operating under a research exemption and not exposed to public users, the current state is defensible. For any production deployment, it would not be.

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